Real-Only vs 10% GAN in Browser Scenarios

Same architecture family • frozen three-seed ensembles • separately frozen equal-calibration thresholds

RESULT

+10% GAN WINS BOTH DEPLOYMENT-SCENARIO F1 COMPARISONS

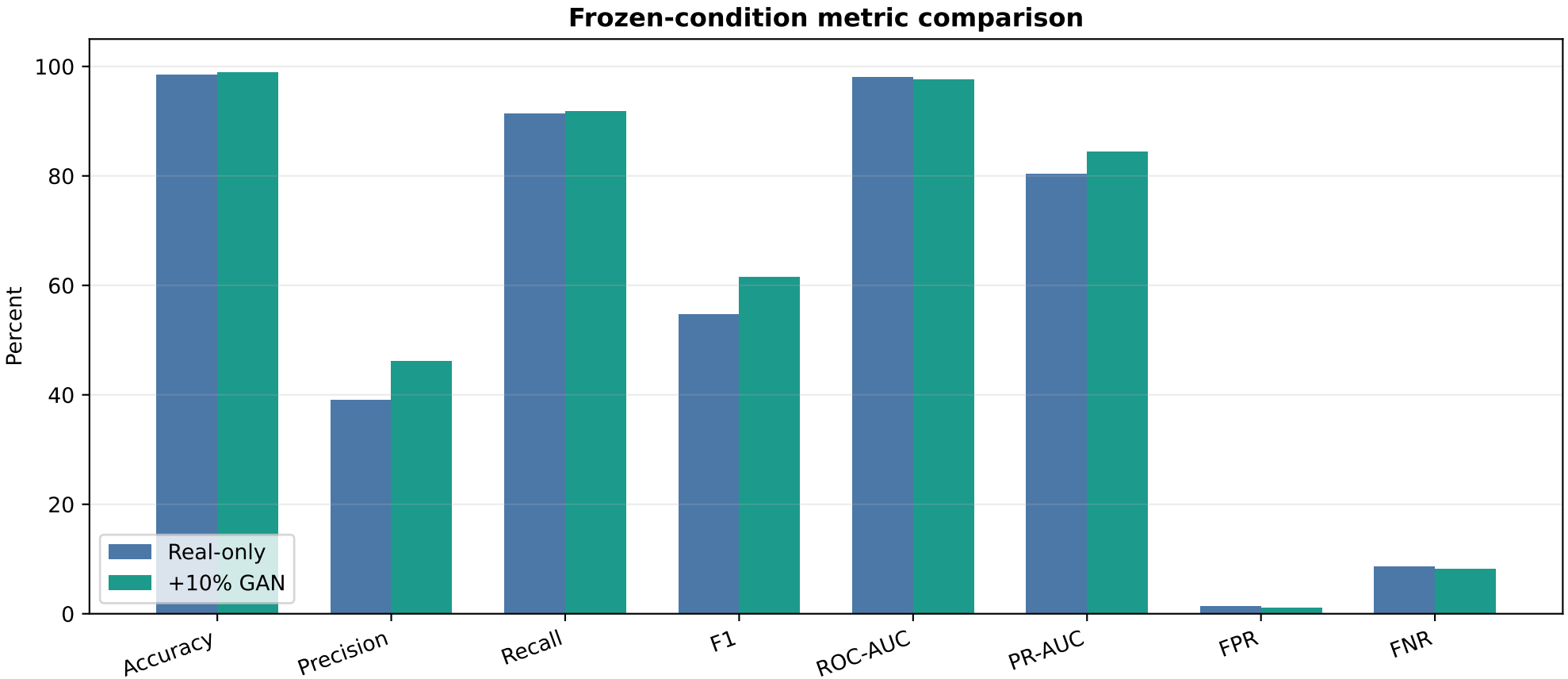
The low-dose GAN-augmented weighted hybrid is compared against its real-only counterpart with the same architecture family, seed ensemble, external datasets, and equal-calibration protocol. Thresholds remain the separately frozen values selected before these scenarios.

Scenario	Condition	Acc.	Prec.	Recall	F1	AUC	PR-AUC	FPR	FNR
0.99% phishing	Real-only	98.50%	39.06%	91.40%	54.73%	98.03%	80.34%	1.43%	8.60%
0.99% phishing	+10% GAN	98.86%	46.22%	91.80%	61.49%	97.62%	84.38%	1.07%	8.20%
5.00% phishing	Real-only	98.03%	76.44%	87.60%	81.64%	97.29%	88.40%	1.42%	12.40%
5.00% phishing	+10% GAN	98.37%	79.91%	89.90%	84.61%	97.13%	91.22%	1.19%	10.10%

No weights or thresholds were changed. The 10% condition refers to GAN augmentation during training—not 10% phishing prevalence.

Scenario: 0.99% phishing

50,500 unseen URLs • 50,000 legitimate • 500 phishing

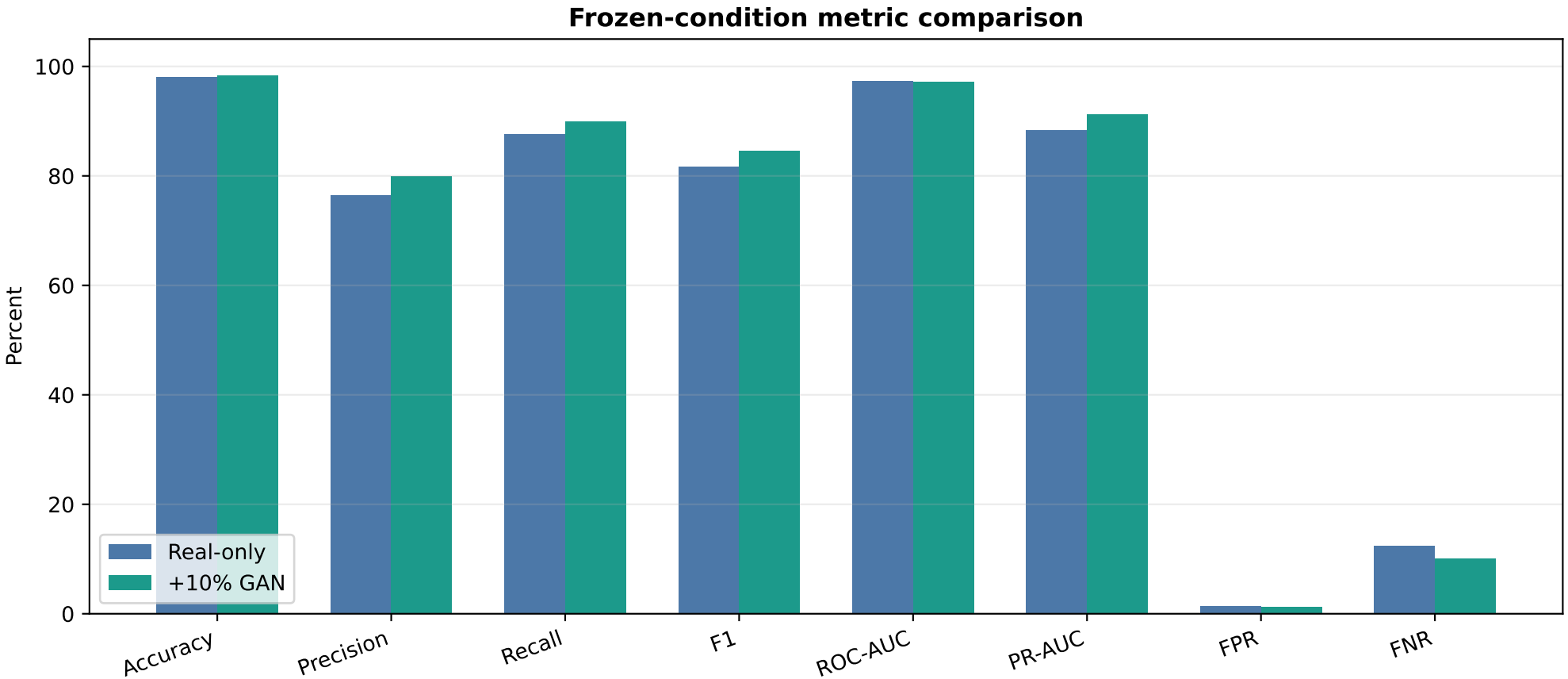


GAN − real-only: F1 +6.76 points; recall +0.40; precision +7.16; ROC-AUC -0.41; FPR -0.36 points.

Positive deltas favor GAN for F1/recall/precision/AUC; a negative delta favors GAN for FPR/FNR.

Scenario: 5.00% phishing

20,000 unseen URLs • 19,000 legitimate • 1,000 phishing



GAN − real-only: F1 +2.97 points; recall +2.30; precision +3.47; ROC-AUC -0.16; FPR -0.23 points.

Positive deltas favor GAN for F1/recall/precision/AUC; a negative delta favors GAN for FPR/FNR.

Operational Counts and Decision

Prevalence-sensitive alert burden

Scenario	Condition	TN	FP	FN	TP	False alerts / 10K	True alerts / 10K
0.99% phishing	Real-only	49,287	713	43	457	141.2	90.5
0.99% phishing	+10% GAN	49,466	534	41	459	105.7	90.9
5.00% phishing	Real-only	18,730	270	124	876	135.0	438.0
5.00% phishing	+10% GAN	18,774	226	101	899	113.0	449.5

INTERPRETATION

- Use F1, recall, FPR, and ROC-AUC together; precision changes substantially with phishing prevalence.
- Any gain belongs to the complete +10% training condition, not automatically to GAN novelty alone.
- The next honest test would require a new untouched temporal cohort; do not tune either threshold on these two datasets.

Because both deployment cohorts have now been used for model comparison, they are retrospective evidence and must not be reused for further threshold tuning.